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Late Cretaceous to early Paleogene paleomagnetic results from Sikhote Alin, far eastern Russia: implications for deformation of East Asia

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Abstract

Welded tuffs in the Bogopol and Sijanov groups were sampled at 27 sites from 12 caldera formations in the Sikhote Alin mountain range around Kavalerovo (44.3°N, 135.0°E) for chronological and paleomagnetic studies. K–Ar age dates show that the welded tuffs erupted between 66 Ma and 46 Ma. All sites yield reliable paleomagnetic directions, with unblocking temperatures higher than 560°C. The high-temperature component at 12 sites and the medium-temperature component at 3 sites in the Bogopol Group show reversed polarity ($D = 193.7°$, $I = -57.6°$, $\alpha_{95} = 8.1°$). The high-temperature component at 11 sites in the Sijanov Group showed both reversed and normal polarities and its mean direction reveals no detectable deflection from north ($D = -2.9°$, $I = 59.6°$, $\alpha_{95} = 11.2°$). The combined paleomagnetic direction of the two groups yields a paleomagnetic pole of 250.5°E, 84.1°N ($A_{95} = 8.8°$), which falls near Cretaceous paleomagnetic poles from Outer Mongolia, Inner Mongolia, the North China Block and the South China Block. The Sikhote Alin area appears not to have been subjected to detectable motion with respect to East Asia since about 50 Ma. This implies that the Sikhote Alin area behaved as an integral part of East Asia during the opening of the Japan Sea at about 15 Ma. However, significant separation between the paleomagnetic poles of East Asia and Europe during the Jurassic–Paleogene implies a major relative movement between these two blocks since the Paleogene.

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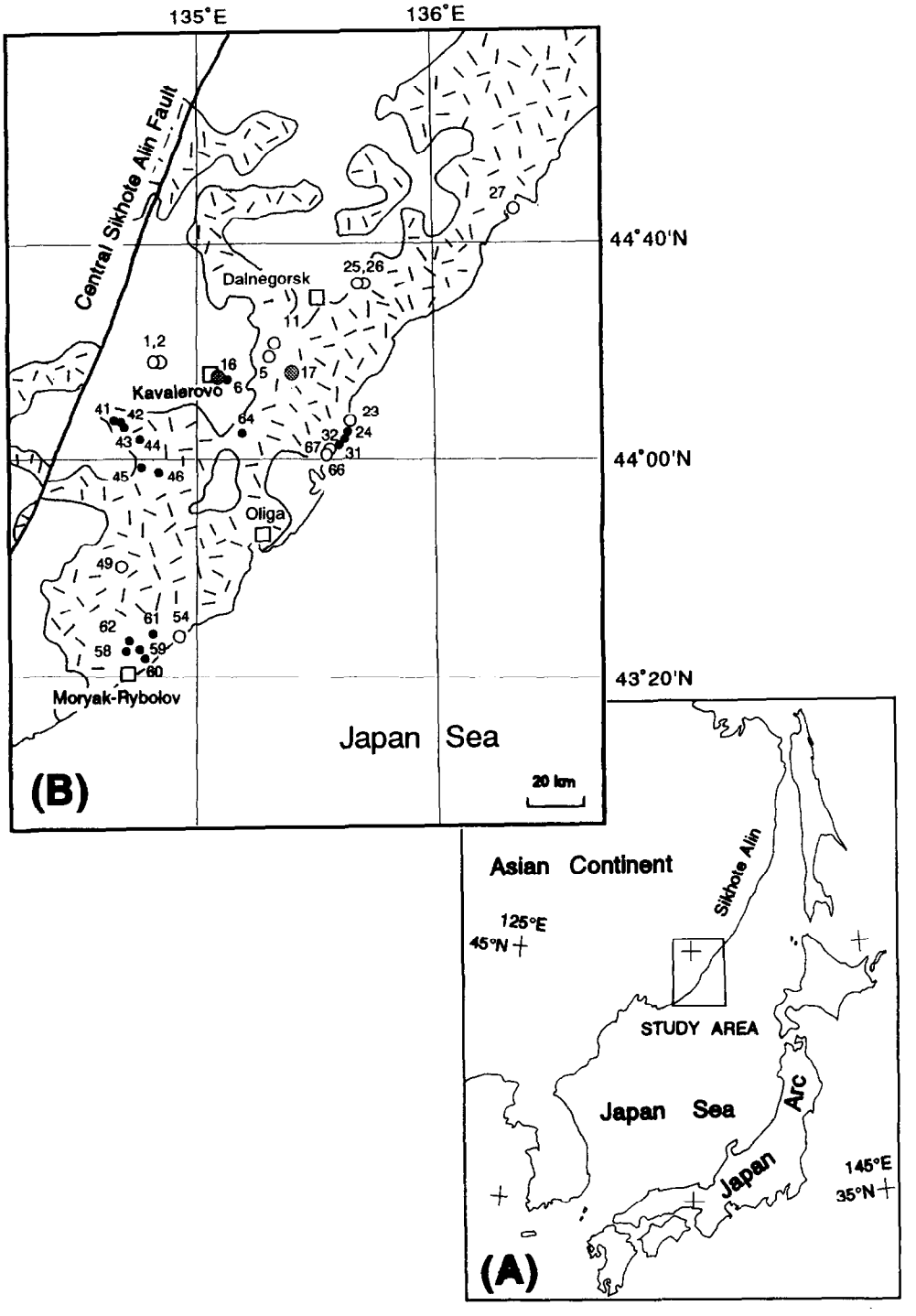


Fig. 1. (A) Sketch map of the study area and configuration of the eastern margin of Asia around the Japan Sea–Japan arc system. (B) Map of study area of Sikhote Alin showing location of paleomagnetic and chronological sampling sites. ● = the Bogopol Group; ○ = the Sijanov Group; shaded circle = chronological sampling sites; stippled area = the distribution of Late Cretaceous to early Paleogene welded tuffs.

1. Introduction

Many island arc–back arc basin systems are distributed in the transition zone between the Eurasian continent and the Pacific Ocean, from the Kuril Basin–Kuril Islands system southward to the South China Sea–Palawan arc system. The opening process of back arc basins is still a subject of controversy. Laboratory and numerical experiments predict that back arc basin formation is a consequence of continental deformation: the collision of India produced contemporaneous tectonic events, varying between continental deformation and back arc spreading, along the eastern margin of the Eurasian continent (e.g., [1–3]).

Recent geological and geophysical studies suggest that the opening of back arc basins along the Eurasian continental margin has been in progress since the India–Asian collision occurred at about 45 Ma: the Japan Sea, the Okinawa Trough and the South China Sea were created later than 32 Ma [4–7]. The opening of the South China Sea was accompanied by large, continental, tectonic rotation of the Indochina block and left-lateral strike-slip motion along the Red River fault system [6]. Hence, the opening process of back arc basins seems to be associated with deformation of the continental margin as a consequence of the ongoing impingement of the Indian subcontinent into Asia.

Formation of the Japan Sea–Japan arc system is one of the major tectonic events along the Eurasian continental coast. The Sikhote Alin mountain area is located in the easternmost part of the Eurasian continental margin, bordering the northern Japan Sea (Fig. 1A). This study focuses on the Sikhote Alin area and uses paleomagnetism to investigate continental deformation associated with the Japan Sea opening. In this paper we present paleomagnetic directions of the Cretaceous to Paleogene welded tuff sequences, which are more reliable than the previous work on lavas of this region [8], and interpret the data in terms of tectonic behavior of the Sikhote Alin area since the Cretaceous.

2. Geological setting and sampling

The Sikhote Alin belt is located east of the Khingan–Bureya and Khankay massifs and comprises thick, Lower Cretaceous flysch deposits and volcanic rocks of Late Cretaceous to Paleogene age [9]. The Sikhote Alin belt is cut by a system of left-lateral strike-slip faults. Movements on these faults were either contemporaneous with, or slightly earlier than, the Late Cretaceous volcanic activity [9].

Extensive volcanic activity accompanied by emplacement of plutons occurred in this belt from 80 Ma to 50 Ma. The volcanic rocks are mostly acid to intermediate welded pyroclastics with interleaved lavas and clastic rocks. According to the definitions of Matyunin [10] and Sakhno et al. [11], the Sikhote Alin volcanic belt is composed of six volcanic groups (Table 1), in ascending order: the Sinancha, Kisin, Sijanov (I, II and III), Bogopol, Tadushi and Suvorov groups. Welded tuffs erupted during formation of the Kisin, Sijanov and Bogopol groups. Rhyolitic welded tuffs of the Kisin Group are the earliest ignimbrites in the volcanic activity and form voluminous and extensive shields in Sikhote Alin [11]. Andesitic to rhyolitic welded tuffs of the Sijanov Group and rhyolitic welded tuffs of the Bogopol Group are ignimbrites associated with later stage large caldera formation. Waning volcanic activity continued into the Eocene and produced the Suvorov basalts.

Welded tuffs of the Bogopol and Sijanov volcanic groups were collected as paleomagnetic samples around Kavalerovo (44.3°N, 135.0°E) (Fig. 1B). More than 160 samples of the Bogopol Group were obtained from 16 sites in six caldera complexes (the Cornorechenskaya, Tumanovskaya, Pavlovskaya, Renin Pass, Margaritovskaya and Petropavlovskaya calderas) and 110 samples of the Sijanov Group were collected at 11 sites in six caldera complexes (the Luzhkinskaya, Siyanovskaya, Yakutinskaya, Dalnegorskaya, Plastunskaya and Tumanovskaya calderas). Hand samples were taken and a magnetic compass was used for orientation. Independently oriented

Table 1
Stratigraphy of the Sikhote Alin belt

Group	Lithology and sampling site name
Voznov	Siltstones
Svetlaja	Molasse
Suvorov (45.8 Ma)	Basaltic lavas SA 17
Tadushi	Volcanic sediments
Bogopol (50.6 Ma, 52.6 Ma)	Rhyolitic welded tuffs
	SA 6, SA 24, SA 31, SA 32, SA 41, SA 42, SA 43, SA 44, SA 45, SA 46, SA 47, SA 58, SA 59, SA 60, SA 61, SA 62, SA 64
Sijanov:	
Samarga (66.0 Ma)	Basaltic lavas SA 16
III	Andesitic welded tuffs
II	Andesitic welded tuffs
	SA 5, SA 9, SA 10, SA 11, SA 25, SA 26, SA 27, SA 66, SA 67
I	Rhyolitic welded tuffs
	SA 1, SA 2, SA 23, SA 49
Kisin	Rhyolitic welded tuffs
Sinancha	Basaltic lavas

samples at each site were distributed over distances ranging up to 20 m. Bedding planes were carefully observed for all the sampling sites on the basis of eutaxitic structure (lineation of stretched pumice and aligned phenocrysts). The bedding tilts were fairly small, varying from 4° to 29°: the mean dip was 13° ± 7°. The present geomagnetic field declination value at each sampling point was evaluated using the International Geomagnetic Reference Field of 1991 [12].

Four samples were collected for K–Ar dating. Two samples were from lavas in the Sijanov Group (Samarga basalts: SA16) and in the Suvorov Group (Suvorov basalts: SA17), the other samples were from the welded tuffs at sites SA31 and SA32 in the Bogopol Group.

3. K–Ar dating

The K–Ar dating was conducted for whole rock samples of the Samarga and Suvorov basalts (SA16 and SA17) and for biotite separates from the Bogopol welded tuffs (SA31 and SA32). Thin sections of samples were examined to determine their suitability for K–Ar dating. The basalts and welded tuffs were found to be virtually free from alteration.

Samples for K–Ar analysis were prepared using the method described by Briggs et al. [13]. Groundmass-rich separates of 80–100 mesh size of basalt samples and the biotite separates of 60–100 mesh size of tuff samples were used for the argon analysis, a portion of each of the separates was further ground to use for potassium analysis. Analyses of potassium and argon, and calculation of ages and errors, were carried out according to the method described by Nagao et al. [14] and Itaya et al. [15]. The decay constants of Steiger and Jäger [16] were used in the age calculations.

K–Ar age data are presented in stratigraphic sequence in Table 2. Detailed stratigraphic locations are given in Table 1 and show good agreement of the K–Ar dates with stratigraphic relationships: the age of the Samarga basalt in the Sijanov groups, the oldest volcanic rocks in our study, is 66.0 Ma, close to the Cretaceous–Tertiary boundary; the Bogopol welded tuffs yield two age dates of 50.6 Ma and 52.6 Ma, whilst the uppermost Suvorov basalts yield an age of 45.8 Ma. These ages are in broad agreement with the

Table 2
K–Ar dates of volcanic rocks of Sikhote Alin

Group	site	Sample	K-Ar age (Ma)	Potassium (wt. %)	Rad. argon 40 (10-8ccSTP/g)	Non rad. Ar (%)	Locality. Lat.(°N)	Long.(°E)
Suvorov	SA 17	Whole rock of basalts	45.8±1.1	0.96±0.02	173.6±2.6	30.6	44.27	135.39
Bogopol	SA 31*	biotite from welded tuffs	50.6±1.2	2.94±0.06	585.6.±6.8	13.3	44.07	135.62
Bogopol	SA 32*	biotite from welded tuffs	52.6±1.3	2.53±0.05	524.4±7.7	28.6	44.05	135.62
Samarga	SA 16	Whole rock of basalts	66.0±2.0	1.14±0.02	298.3±6.8	49.7	44.26	135.06

Ages were calculated from the following decay and abundance constants: $\lambda\epsilon = 0.581 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}; $40K/K = 1.167 \times 10^{-4}$ mol mol^{-1} [16]. SA 31* and SA 32*: K data show significantly low potassium content in biotite separates, suggesting that the potassium in biotite phenocrysts was leached during the exposure time of the host rocks. Generally, however, the leaching process does not affect any significant K–Ar age [44]: similar K–Ar ages were observed for biotite and hornblende separates as well as plural biotite separates with different K contents from the same stratigraphic sequence.

paleontologically assigned age of Late Cretaceous–Paleogene [10].

4. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 22 mm long, were prepared from samples for each site. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer or a ScT cryogenic magnetometer, according to intensity of the magnetization. All specimens were subjected to stepwise thermal demagnetization up to 630°C. Results for each specimen were plotted on orthogonal vector diagrams [17] to recognize components constituting the NRM, as well as on equal-area projections. Principal component analysis [18] was used to estimate the direction.

4.1. Bogopol welded tuffs

Two magnetization components are observed in 12 sites (e.g., Fig. 2A), whereas the remaining four sites (SA58, SA59, SA60 and SA62), from the Margaritovskaya Caldera, show at least three magnetization components (Fig. 2B). The low-temperature components are removed from all sites in the initial stage of demagnetization by 300°C. In situ directions of the low-temperature component are close to the geocentric axial dipole field and are probably thermoviscous magnetization acquired in the recent geomagnetic field [19]. The high blocking temperature component is distinctly separated and is unblocked between 590°C and 620°C, indicating that this component resides both in magnetite and hematite. Circles of 95% confidence of the site mean directions of the high-temperature component do not exceed 15° (Table 3).

A southerly direction with reversed polarity was isolated from 12 sites in the high-temperature component. Mean directions before and after tilt correction are similar ($D = 190.4°$, $I = -57.1°$ vs. $D = 193.2°$, $I = -55.8°$). The precision parameter (k) of the group mean is slightly increased after correction for bedding tilt ($k_{\text{before}} = 17.0$ vs. $k_{\text{after}} = 26.4$). The improvement is not significant at the 95% confidence level [20,21] but indicates that the remanent magnetization was possibly acquired prior to the tilting.

Three sites from the Margaritovskaya Caldera (SA58, SA59 and SA60) and one site from the Petropavlovskaya Caldera (SA61) reveal an easterly direction with a declination of about 90° in the high-temperature component. Declination

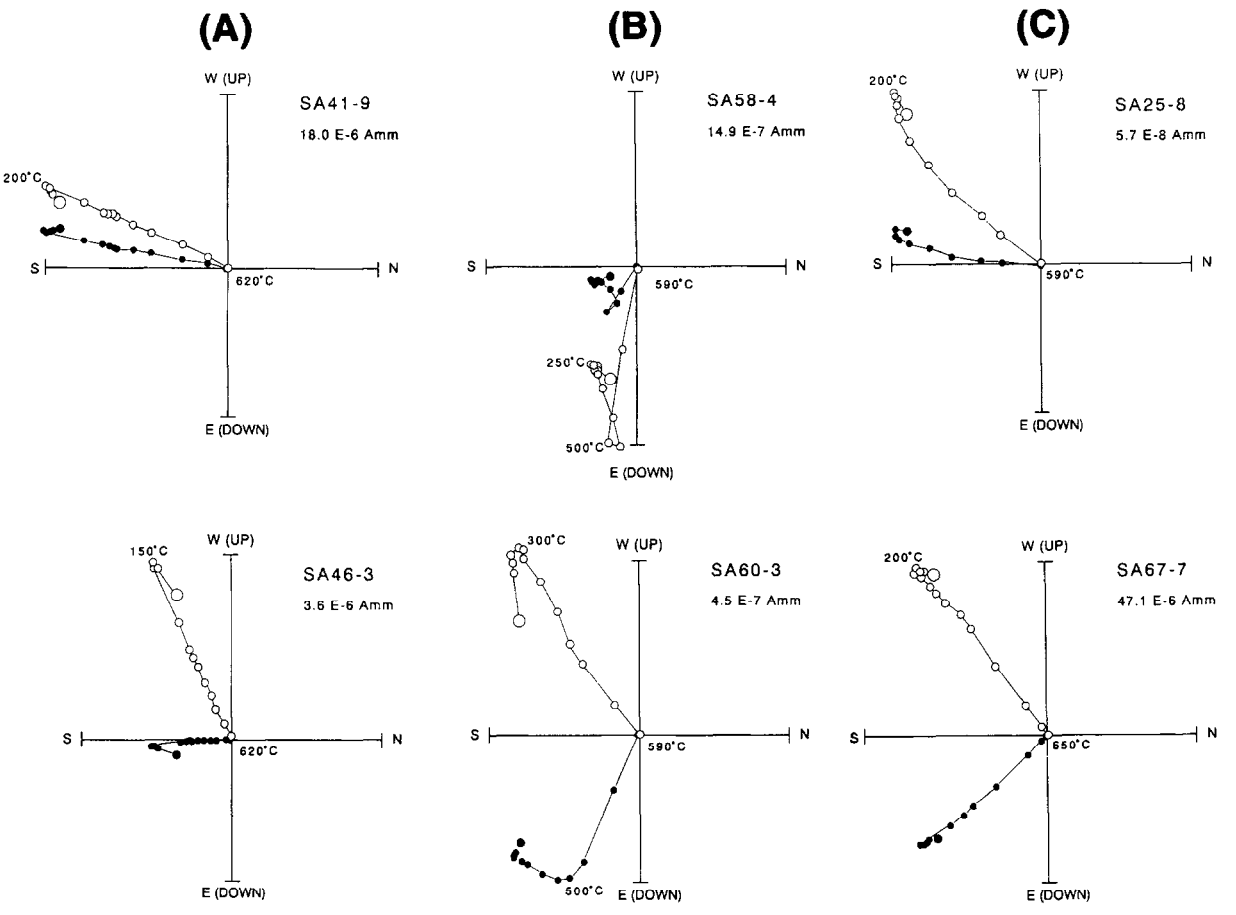


Fig. 2. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments. (A) The Bogopol Group with two-magnetization component. (B) The Bogopol Group with three-magnetization component. (C) The Sijanov Group. ○ = the magnetization vector projection on to the vertical plane; ● = projection on to the horizontal plane. Tectonic tilt correction has not been applied.

ranges over a fairly limited range, from 56° to 101°, whereas inclination is distributed over a wide range, from +51° to −49°.

These anomalous directions of the high-temperature component follow a coherent reversed polarity direction of the middle-temperature component (Fig. 2B). Tightly clustered directions with reversed polarity were successfully isolated at a temperature range between 200°C and 500°C in three sites from the Margaritovskaya Caldera (SA58, SA59 and SA60). Microscopic observation showed that no remarkable petrological evidence for hydrothermal alteration was observed in samples of SA59, SA60 and SA61, whereas a sample of SA58 was weakly altered. This indicates that the high-temperature component with the anomalous direction, as well as the middle-temperature component with the reversed direction, are not attributed to chemical remanent magnetization, with a physicochemical cause, but is rather ascribable to thermoremanent magnetization (TRM). Anomalous geomagnetic field behavior (e.g., excursion) recorded in the high-temperature component only lasted until the welded tuffs cooled to about 500°C. The reversed direction is possibly partial TRM, acquired during initial cooling of the welded tuffs below about 500°C.

Because the mean direction of the 3 sites

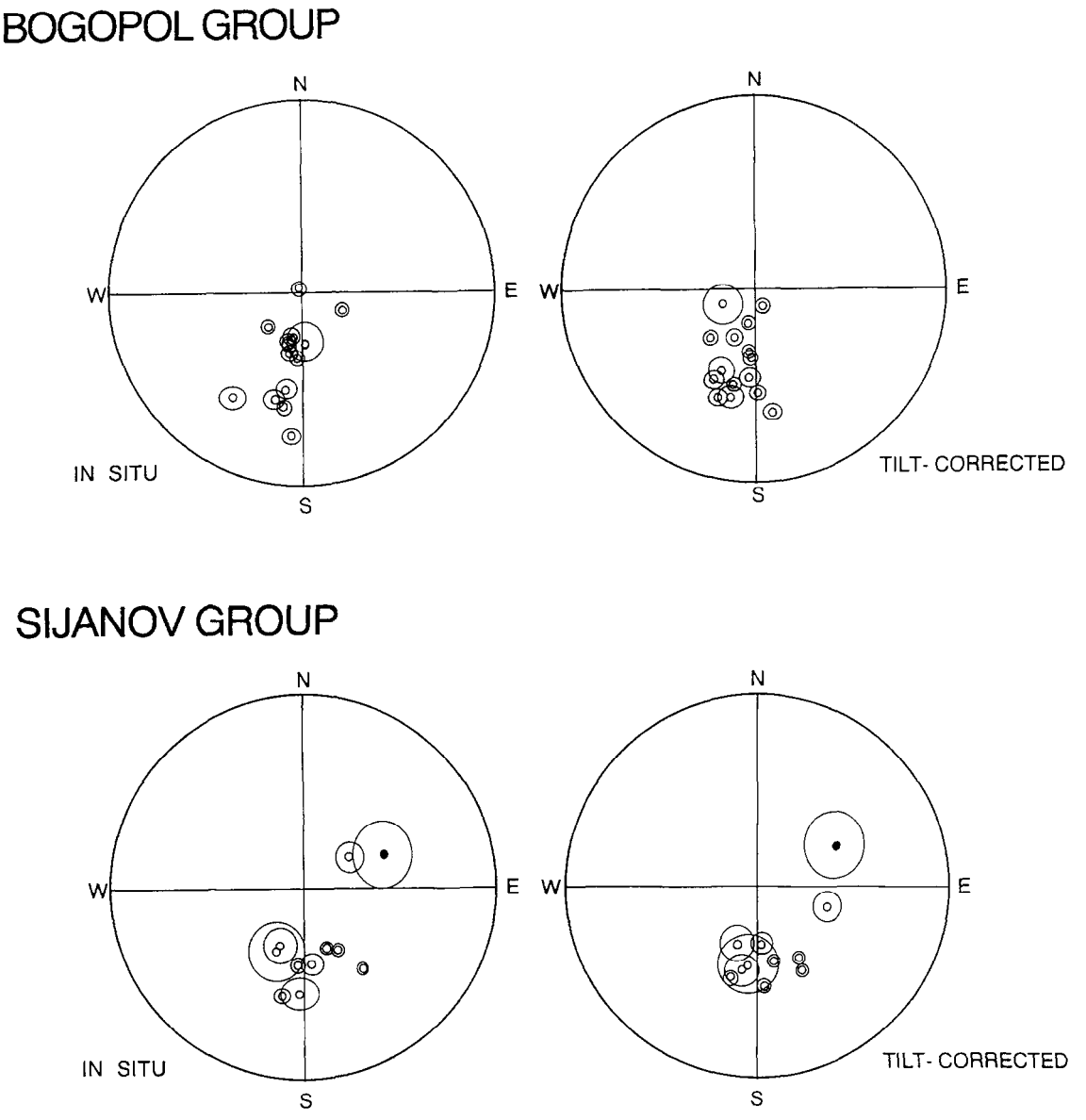


Fig. 3. Equal-area projections of the site mean direction with a 95% confidence circle for the Bogopol Group and the Sijanov Group, before and after tilt correction. Data of site SA61 are not shown in this figure. ● = the lower hemisphere; ○ = the upper hemisphere. Fisherian precision parameter, k, and the radius of cone of 95% confidence, α_{95}, changes from $k = 16.5$, $\alpha_{95} = 9.7°$ to $k = 23.5$, $\alpha_{95} = 8.1°$ for the Bogopol Group, and from $k = 12.8$, $\alpha_{95} = 13.3°$ to $k = 17.7$, $\alpha_{95} = 11.2°$ for the Sijanov Group after tilt correction.

exhibiting the middle-temperature component ($D = 196.1°$, $I = -64.3°$) after tilt correction is compatible at the 95% confidence level with the direction of the high-temperature component of the 12 sites with reversed polarity, these three data are incorporated into the remaining reversed data to calculate an overall mean direction for the Bogopol Group of $D = 193.7°$, $I = -57.6°$

Table 3
Summary of paleomagnetic results for the Late Cretaceous to Paleogene welded tuffs from Sikhote Alin, Russia

SITE	DEMAG.(°C) COMPONENT	N	IN SITU D(°)	IN SITU I(°)	TILT CORRECTED D(°)	TILT CORRECTED I(°)	k	α95(°)	LOCALITY Lat.(°N)	LOCALITY Long.(°E)
Bogopol Group (Rock Type: rhyolitic welded tuffs)										
[Cornorechenskaya caldera]										
SA06	HT 200-590	10	-170.1	-47.8	-166.3	-41.9	104.6	4.7	44.47	135.70
[Tumanovskaya caldera]										
SA24	HT 250-620	10	-164.6	-67.0	-174.6	-63.1	951.5	1.6	44.10	135.63
SA31	HT 150-620	10	-168.1	-63.3	-176.1	-60.7	228.7	3.2	44.07	135.62
SA32	HT 200-620	10	-175.5	-62.7	179.3	-45.1	210.1	3.3	44.05	135.62
[Pavlovskaya caldera]										
SA41	HT 100-620	10	-175.7	-26.6	172.7	-36.2	187.5	3.5	44.10	134.68
SA42	HT 100-620	10	-145.9	-35.1	-156.3	-52.0	95.9	5.0	44.10	134.69
SA43	HT 100-620	10	-170.5	-40.1	-167.0	-48.3	263.4	3.0	44.10	134.69
SA44	HT 150-620	10	-164.8	-42.6	-175.0	-51.6	135.9	4.2	44.06	134.74
SA64	HT 200-620	10	113.0	-71.3	-168.1	-75.0	285.0	2.9	44.08	135.18
[Renin pass caldera]										
SA45	HT 200-620	10	-136.0	-69.8	-137.2	-61.8	226.5	3.2	43.98	134.76
SA46	HT 150-620	10	177.2	-69.1	-115.2	-75.1	36.0	8.2	43.96	134.84
[Margaritovskaya caldera]										
SA58	HT† 500-590	9	100.4	70.7	56.0	51.8	145.5	4.3	43.40	134.70
	MT* 200-400	9	-167.2	-69.0	-160.8	-40.2	151.8	4.2		
SA59	HT† 500-590	10	74.8	-51.5	86.3	-48.6	13.9	13.4	43.41	134.76
	MT* 200-500	7	-26.3	-87.4	153.9	-82.5	369.5	3.1		
SA60	HT† 500-590	10	99.6	-30.0	101.2	-34.3	194.1	3.5	43.40	134.76
	MT* 250-500	10	-167.3	-71.0	-156.8	-67.9	152.2	3.9		
SA62	HT 450-590	9	177.6	-68.5	-154.6	-46.8	152.8	4.2	43.44	134.71
[Petropavlovskaya caldera]										
SA61	HT† 100-620	10	73.1	-26.0	79.0	-37.8	193.2	3.5	43.46	134.81
Bogopol Group										
Mean	HT	12	10.4	57.1	13.2	55.8	26.4	8.6	44.00	135.00
Mean	HT+MT*	15	10.8	61.3	13.7	57.6	23.5	8.1		
Sijanov II Group (Rock type: andesitic to rhyolitic welded tuffs)										
[Siyanovskaya caldera]										
SA05	HT 300-590	8	-157.8	-63.7	-161.6	-64.5	59.9	7.2	44.33	135.28
SA11	HT 150-590	9	149.8	-60.5	151.0	-49.5	553.8	2.2	44.35	135.32
[Dalnegorskaya caldera]										
SA25	HT 200-590	9	-177.7	-44.5	-169.3	-54.1	53.5	7.1	44.53	135.66
SA26	HT 200-590	10	-168.4	-43.3	-163.6	-49.9	268.5	3.0	44.53	135.67
[Plastunskaya caldera]										
SA27	HT 150-620	10	67.2	52.6	62.2	51.6	13.9	13.4	44.77	136.30
[Tumanovskaya caldera]										
SA66	HT 100-590	10	-175.2	-57.8	176.0	-47.7	363.6	2.5	44.04	135.58
SA67	HT 200-650	10	142.6	-47.4	149.0	-54.8	471.7	2.2	44.05	135.59
Sijanov I Group (Rock type: andesitic to rhyolitic welded tuffs)										
SA49	HT 100-650	10	-156.8	-60.8	-173.6	-56.6	16.5	12.3	43.66	134.67
[Luzhkinskaya caldera]										
SA01	HT 150-590	9	159.8	-63.0	166.2	-58.2	492.2	2.3	44.30	134.82
SA02	HT 300-620	10	173.9	-57.7	175.7	-65.7	97.1	4.9	44.30	134.83
[Tumanovskaya caldera]										
SA23	HT 500-590	10	56.7	-66.7	105.5	-58.9	65.8	6.0	44.12	135.63
Sijanov II and Sijanov I Groups										
Mean	HT	11	-0.6	62.0	-2.9	59.6	17.7	11.2	44.00	135.00
Bogopol and Sijanov Groups (mean)										
	HT	23	5.6	59.5	6.0	57.9	20.9	6.8	44.00	135.00
	HT+MT*	26	6.1	61.7	7.0	58.7	20.3	6.5		

($\alpha_{95} = 8.1°$) (Fig. 3). Grouping of the direction increases from $k_{before} = 16.5$ to $k_{after} = 23.5$, after tilt correction.

4.2. Sijanov Group

Eleven sites exhibited two distinct magnetization components. The high unblocking temperature component was isolated above 300°C (Fig. 2C) and unblocked in the temperature range 590–650°C, indicating residence in both magnetite and hematite, as found in the Bogopol Group.

The high-temperature component from 10 sites showed reversed polarity and 1 site exhibited normal polarity (Fig. 3). Paleomagnetic directions of 11 sites clustered more tightly after tilt correction ($k_{before} = 12.8$ vs. $k_{after} = 17.7$), indicating that the remanent magnetization was possibly acquired before tilting occurred. The tilt-corrected group mean direction is $D = -2.9°$, $I = 59.6°$ ($\alpha_{95} = 11.2°$), which is not significantly deflected from north.

5. Discussion

5.1. Late Cretaceous–Paleogene paleomagnetic pole for Sikhote Alin

Reliable reversed polarity directions appear from the high- and middle-temperature remanent magnetization components at 25 sites in the Bogopol and Sijanov groups. The K–Ar results assign ages of 51 and 53 Ma to the former group, and 66 Ma to the latter group. Reversed polarity is dominant during the period between 66 Ma and 51 Ma [22]. A primary origin for their remanent magnetization was ascertained by the predominance of the reversed polarity, the stable behavior under thermal demagnetization and a closer grouping of the paleomagnetic directions after tilt correction.

A Late Cretaceous–early Paleogene pole was evaluated on the basis of data set of both the Sijanov and the Bogopol groups. Since formation ages for the Bogopol and Sijanov groups cover a fairly long duration, between 51 Ma and 66 Ma, the secular variation is almost certainly averaged out. The pole falls at a position of latitude 84.1°N, longitude 250.5°E ($A_{95} = 8.8°$). The paleomagnetic direction at a representative point of the sample location (135°E, 44°N) is $D = 7.0°$, $I = 58.7°$ ($\alpha_{95} = 6.5°$), close to a result previously obtained by Bretstein [8] from Late Cretaceous–early Paleogene rocks of this region, although no tilting corrections were applied in the latter study. Results of the present study are a more reliable determination of the Late Cretaceous–Early Paleogene magnetic field direction and the paleomagnetic pole calculated from the two groups is recognized as a representative of the entire Sikhote Alin area because of the wide distribution of the rocks sampled.

5.2. Comparison of the Sikhote Alin pole with Cretaceous poles for East Asia

Fig. 4 shows the Cretaceous poles for Sikhote Alin and neighboring regions. Zhao et al. [23] and Pruner et al. [24] obtained Cretaceous poles from Inner Mongolia (Lat. 82.9N°, Long. 249.5°E) and Outer Mongolia (Lat. 82.9°N, Long. 241.7°E), respectively (Table 4). The North China pole is recalculated from data from [25–28], compiled by Zhao et al. [23] and Zheng et al. [28]: Lat. 86.9°N, Long. 255.7°E ($A_{95} = 9.2°$). The Sikhote Alin pole is found to be in good agreement with poles from

Notes to Table 3:
Magnetization directions of high (HT) and middle (MT*) temperature components are listed. N = number of samples in calculation for the means; °C = the temperature range used to calculate characteristic direction by principal component analysis; D and I = declination and inclination, respectively; k = the estimate of the Fisherian precision parameter; α_{95} = the radius of cone of 95% confidence. The mean direction of the high-temperature component for the Bogopol Group (HT) was calculated from 12 sites, except for the following four sites: SA 58, SA 59, SA 60 and SA 61 (HT†). The middle-temperature component directions from three sites (MT*; SA 58, SA 59 and SA 60) were combined to calculate a mean direction (HT + MT*) of the Bogopol Group.

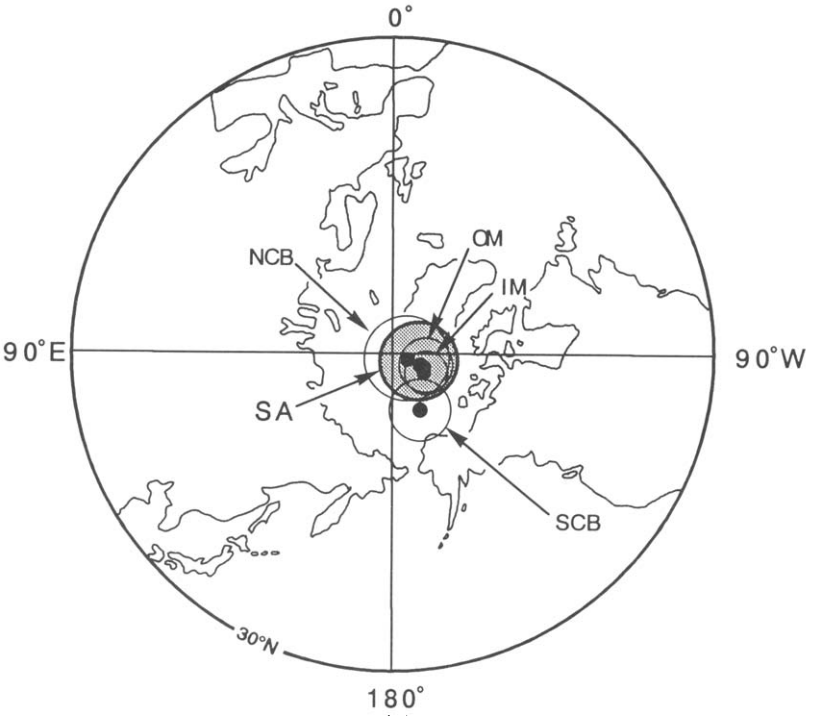


Fig. 4. Cretaceous paleomagnetic poles (with associated circles of 95% confidence) for Sikhote Alin (shaded) and some other parts of East Asia. *SA* = Sikhote Alin; *IM* = Inner Mongolia; *OM* = Outer Mongolia; *NCB* = the North China Block; *SCB* = the South China Block.

Inner Mongolia, Outer Mongolia and North China: poles for Sikhote Alin and the neighboring regions are indistinguishable within the 95% confidence limit. The polar distances between the Sikhote Alin pole and other poles is less than 2.8°.

A noticeable differences exists, however, between the Late Cretaceous poles for the South China block (Lat. 77.2°N, Long. 205.9°E, A_{95} = 6.3°) and Sikhote Alin. The South China pole is recalculated from the data set of six areas [29–33]. Although deviation of the pole for Sikhote Alin from the South China pole is 9.5° (±10.8°), two circles of 95% confidence just overlap (Fig. 4).

This comparison of poles suggests that Sikhote Alin has been subjected to little or no rotational motion and north–south movement relative to the eastern margin of the Asian continent since the Late Cretaceous. Because the left-lateral strike-slip fault bordering Sikhote Alin had ceased motion by the Late Cretaceous [9], it appears that deformation due to its displacement had ceased by the time that the Sijanov and Bogopol groups were formed, about 60 Ma. Hence, Sikhote Alin has behaved as an integral part of East Asia at least since the Late Cretaceous.

The present study implies that large tectonic displacement of Sikhote Alin did not occur during the opening of the Japan Sea. Recent geological and geophysical studies of the Japan Sea floor show that the Japan Sea opened between 20 Ma and 14 Ma [5]. Paleomagnetic studies from the Japan arc suggest that the synchronous rotational motion of southwest Japan and northeast Japan with respect to East Asia attained a climax at about 15 Ma [7,34]. The present study has not succeeded in detecting intracontinental deformation in East Asia since 52 Ma.

This stable aspect of the continental margin in eastern Asia contrasts with the evolution of the South China Sea and terranes further south. Rotation of Indochina and Borneo lead to the open-

ing of the South China Sea as these blocks were extruded towards the southeast [1,35]. Deformation of the continent, associated with strike-slip tectonics, has occurred in Indochina, as well as along the arc side, in the South China Sea–Palawan arc system. In the Japan Sea–Japan arc system, however, large tectonic events are not observed along the continental side: the Japan Sea apparently opened smoothly without deformation of the continental margin. This implies that the opening event of the Japan Sea was decoupled from the continental crust deformation during the extrusion tectonics claimed by Tapponnier et al. [1]. The opening of the Japan Sea and rifting of the Japan arc may be induced from the flow of asthenospheric material without significant intracontinental deformation [36,37].

5.3. Relative motion between East Asia and Europe

Paleomagnetic results from Sikhote Alin provide evidence for the relative rotation between East Asia and Europe (Table 4). The Cretaceous paleomagnetic pole positions of East Asia and Europe are significantly distinguishable at the 95% confidence level (Fig. 5). A combined Cretaceous pole for East Asia from Sikhote Alin, Inner Mongolia, Outer Mongolia, North China and South China falls at 233.5°E, 83.3°N (A_{95} = 4.3°). This position is appreciably to the east of the position from the European Cretaceous pole, which is plotted near 164°E, 74°N [38] or 154°E, 72°N [39]. The polar distance between the East Asian and European poles is more than 15°.

A similar difference is observed in the Jurassic pole positions for Europe and East Asia (Fig. 5): European Jurassic poles are significantly removed from those of China (Table 4). An European pole at 72°N, 162° is given for the Late Jurassic by Van der Voo [39] and an European Jurassic pole (Lat 72°N, Long. 131°E) is calculated from the data set of Besse and Courtillot [40]. A Jurassic pole for the China block at latitude 71.4°N, longitude 217.7°E is given by Enkin

Table 4
Paleomagnetic pole positions from the Jurassic, Cretaceous and Paleogene

[Paleogene] Area	N(n)	Lat.	Long.	A_{95}	References
Sikhote Alin (Bogopol)	(15)	78.9°N	236.7°E	10.3°	This study
Europe (40-60 Ma)	1	75°N	157°E	4°	[40]
Europe (Paleocene/Eocene)	19	76°N	161°E	3°	[42]
[Cretaceous] Area	**N(n)**	**Lat.**	**Long.**	**A_{95}**	**References**
Sikhote Alin (Bogopol+sijanov)	(26)	84.1°N	250.5°E	8.8°	This study
Inner Mongolia	(6)	82.9°N	249.5°E	5.7°	[23]
Outer Mongolia	(3)	82.9°N°	241.7°E	4.2°	[24]
NCB	4	86.9°N	255.7°E	9.2°	[25-28]
SCB	6	77.2°N	205.9°E	6.3°	[29-33]
East Asia†	5	83.3°N	233.5°E	4.3°	This study
Europe (Late Cretaceous*)	5	74°N	164°E	6°	[38]
Europe (Late Cretaceous**)	5	72°N	154°E	6°	[39]
[Jurassic] Area	**N**	**Lat.**	**Long.**	**A_{95}**	**References**
China (Late Jurassic)	5	71.4°N	217.7°E	7.2°	[41]
Europe (Late Jurassic)	7	72°N	162°E	10°	[39]
Europe (147-197 Ma)	6	72.3°N	131.1°E	8.3°	[40]

N = the number of studies or areas; n = the number of sampling localities; A_{95} = the radius of the circle of confidence at the 95% probability level. East Asia†: This pole of East Asia is calculated from data from the following five areas: Sikhote Alin, Inner Mongolia, Outer Mongolia, NCB (North China Block) and SCB (South China Block).

et al. [41]. Thus, the East Asian Jurassic pole also appears to occupy a position to the east of the European poles.

A position for the paleopoles of East Asia to the east of those of Europe is also observed in the Paleogene. Paleomagnetic poles for 40–50 Ma and from the Paleocene–Eocene for Europe are given as (Lat. 75°N, Long. 157°E) and (Lat. 76°N, Long. 161°E), respectively [40,42]. A paleomagnetic pole of 50 Ma for East Asia is repre-

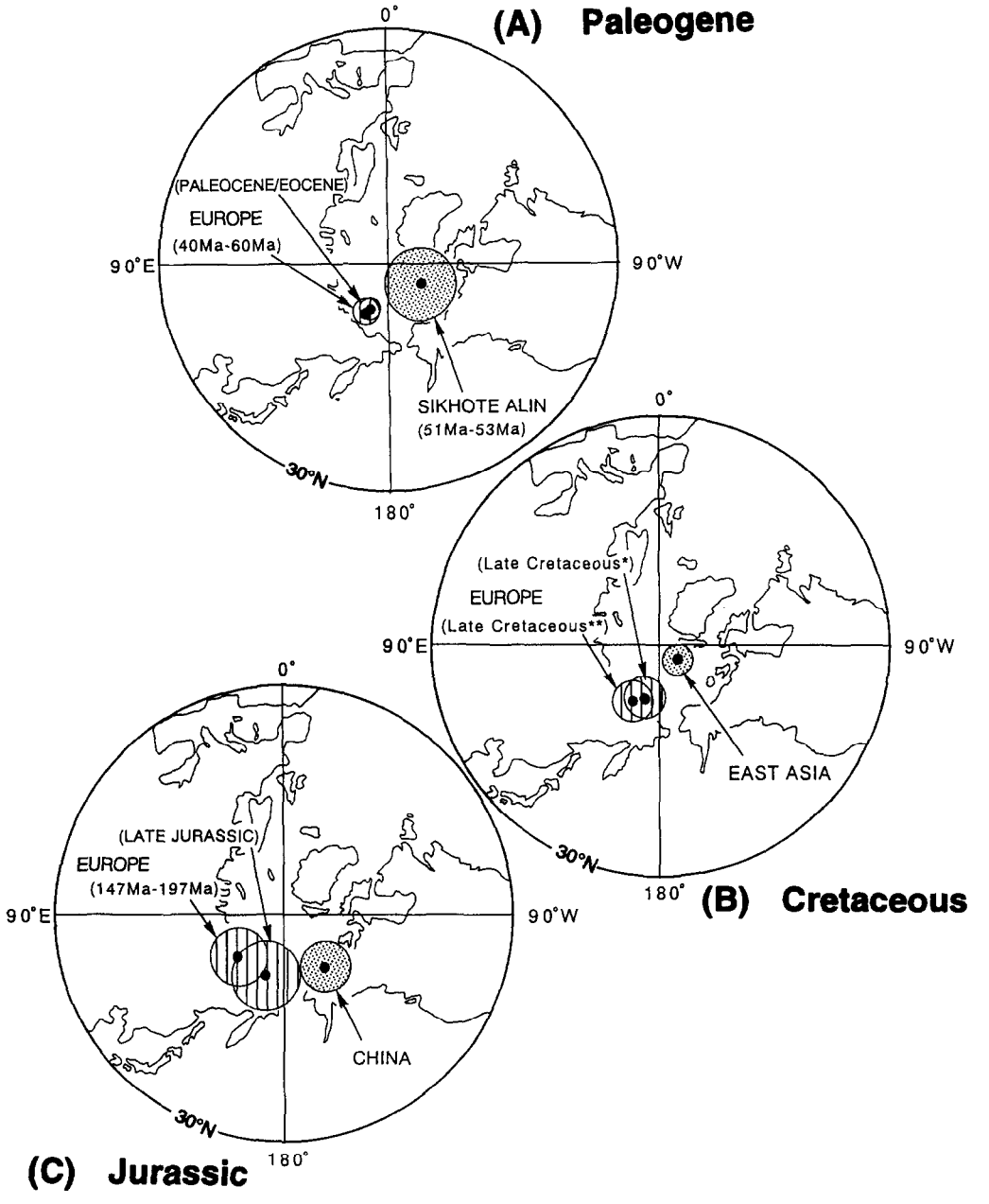


Fig. 5. Paleomagnetic poles (with associated circles of 95% confidence) for East Asia (dotted) and Europe (striped) for three periods: (A) Paleogene; (B) Cretaceous; and (C) Jurassic. Data from Table 4. Polar stereographic projection.

sented by the pole of the Bogopol Group (Lat. 78.9°N, Long. 236.7.°E, $A_{95} = 10.3°$) of Sikhote Alin. The Bogopol pole is located to the east of the European ones and deviates from them by more than 15.5° (±10.7°). Disagreement in the same sense of the pole position between East Asia and Eurasia is therefore recorded during the period between the Jurassic and Eocene.

The arrangement of poles of East Asia and Europe can not easily be ascribed to the effect of northward offset of axial dipole. The amount of deviation of the poles is more than 15°, whereas only 3.9° is expected as a maximum value of northward offset by the dipole-offset theory [43].

The discrepancy in the pole positions between East Asia and Europe is explained by a fairly large relative tectonic motion. This relative movement is described as either a clockwise rotation of Europe without latitudinal movement and/or northward movement of the East Asia. The Sikhote Alin data strongly suggest that relative motion began later than 50 Ma. The Indian continent collided with the Eurasian continent at about 45 Ma and has further impinged into it since then. The large continental deformation due to the collision of India is possibly responsible for the northward movement of East Asia and, hence, for the disagreement of the pole positions of East Asia and Europe. We conclude that the Eurasian continent, extending from Europe to East Asia, did not behave as a rigid continent during the Tertiary.

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